Accelerated Math II/III Name_____

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HW #10 - Rational Root Theorem

Date_____ Period____

State the possible rational zeros for each function.

1) $f(x) = 4x^3 + 7x^2 + 2x - 1$

2) $f(x) = 2x^4 - 27x^3 + 11x^2 + 27x - 13$

3) $f(x) = 4x^4 - 12x^3 - 47x^2 - 42x - 11$

4) $f(x) = 2x^4 + 7x^3 + 4x^2 - 7x - 6$

State the possible rational zeros for each function. Then find all rational zeros. One zero has been given.

5) $f(x) = 2x^4 - x^3 - 11x^2 - 11x - 3;\ 3$

6) $f(x) = 15x^4 + 14x^3 - 12x^2 - 14x - 3;\ -\frac{3}{5}$

State the possible rational zeros for each function. Then find all rational zeros.

7) $f(x) = 2x^3 - 3x^2 - 3x + 9$

8) $f(x) = 4x^3 - 4x^2 - x + 1$

9) $f(x) = 4x^3 + 9x^2 + 6x + 1$

10) $f(x) = 4x^3 + 16x^2 - 25x + 8$

11) $f(x) = 3x^3 - 4x^2 - 5x + 2$

12) $f(x) = 5x^3 + 59x^2 + 43x - 11$

13) $f(x) = 5x^4 - x^3 - 5x^2 + x$

14) $f(x) = 5x^4 + 19x^3 - 29x^2 + 5x$

Describe the end behavior of each function.

15) $f(x) = -x^3 + 2x^2 - 1$

16) $f(x) = -x^4 - x^3 + x^2 + 4$

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Answers to HW #10 - Rational Root Theorem

1) $\pm 1, \pm\frac{1}{2}, \pm\frac{1}{4}$

2) $\pm 1, \pm 13, \pm\frac{1}{2}, \pm\frac{13}{2}$

3) $\pm 1, \pm 11, \pm\frac{1}{2}, \pm\frac{11}{2}, \pm\frac{1}{4}, \pm\frac{11}{4}$

4) $\pm 1, \pm 2, \pm 3, \pm 6, \pm\frac{1}{2}, \pm\frac{3}{2}$

5) Possible rational zeros: $\pm 1, \pm 3, \pm\frac{1}{2}, \pm\frac{3}{2}$

Rational zeros: $\left\{-1 \text{ mult. } 2, -\frac{1}{2}, 3\right\}$

6) Possible rational zeros:

$\pm 1, \pm 3, \pm\frac{1}{3}, \pm\frac{1}{5}, \pm\frac{3}{5}, \pm\frac{1}{15}$

Rational zeros: $\left\{-\frac{1}{3}, 1, -1, -\frac{3}{5}\right\}$

7) Possible rational zeros:

$\pm 1, \pm 3, \pm 9, \pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{9}{2}$

Rational zeros: $\left\{-\frac{3}{2}\right\}$

8) Possible rational zeros: $\pm 1, \pm\frac{1}{2}, \pm\frac{1}{4}$

Rational zeros: $\left\{1, \frac{1}{2}, -\frac{1}{2}\right\}$

9) Possible rational zeros: $\pm 1, \pm\frac{1}{2}, \pm\frac{1}{4}$

Rational zeros: $\left\{-1 \text{ mult. } 2, -\frac{1}{4}\right\}$

10) Possible rational zeros:

$\pm 1, \pm 2, \pm 4, \pm 8, \pm\frac{1}{2}, \pm\frac{1}{4}$

Rational zeros: $\left\{\frac{1}{2}\right\}$

11) Possible rational zeros: $\pm 1, \pm 2, \pm\frac{1}{3}, \pm\frac{2}{3}$

Rational zeros: $\left\{\frac{1}{3}, 2, -1\right\}$

12) Possible rational zeros: $\pm 1, \pm 11, \pm\frac{1}{5}, \pm\frac{11}{5}$

Rational zeros: $\left\{\frac{1}{5}, -11, -1\right\}$

13) Possible rational zeros: $0, \pm 1, \pm\frac{1}{5}$

Rational zeros: $\left\{0, \frac{1}{5}, -1, 1\right\}$

14) Possible rational zeros: $0, \pm 1, \pm 5, \pm\frac{1}{5}$

Rational zeros: $\left\{0, 1, -5, \frac{1}{5}\right\}$

15) $f(x) \to +\infty$ as $x \to -\infty$
$f(x) \to -\infty$ as $x \to +\infty$

16) $f(x) \to -\infty$ as $x \to -\infty$
$f(x) \to -\infty$ as $x \to +\infty$

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